

Numerical Analysis homework V

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I. Detailed proof of Thm

We first check that the operations that $\mathcal{C}[a, b]$ is closed under them.

Closure under addition. Let $f, g \in \mathcal{C}[a, b]$. For each $x \in [a, b]$ we define $(f + g)(x) := f(x) + g(x)$. Since f and g are continuous and the sum of continuous functions is continuous, the function $f + g$ is continuous on $[a, b]$. Hence $f + g \in \mathcal{C}[a, b]$.

Closure under scalar multiplication. Let $f \in \mathcal{C}[a, b]$ and $\alpha \in \mathbb{C}$. For each $x \in [a, b]$ define $(\alpha f)(x) := \alpha f(x)$. The map $x \mapsto f(x)$ is continuous, and multiplication by the constant α is a continuous operation, so the product αf is continuous on $[a, b]$. Thus $\alpha f \in \mathcal{C}[a, b]$.

Now we verify the vector space axioms. All equalities below are understood pointwise on $[a, b]$.

- *Associativity of addition:* For any $f, g, h \in \mathcal{C}[a, b]$,

$$((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x).$$

- *Commutativity of addition:* For any $f, g \in \mathcal{C}[a, b]$,

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x).$$

- *Additive identity:* Let 0 denote the zero function $0(x) \equiv 0$. It is continuous and $(f + 0)(x) = f(x) + 0 = f(x)$ for all x , so $f + 0 = f$.
- *Additive inverse:* For $f \in \mathcal{C}[a, b]$, define $-f$ by $(-f)(x) := -f(x)$. Then $-f$ is continuous and $(f + (-f))(x) = f(x) - f(x) = 0$, so $f + (-f) = 0$.
- *Compatibility of scalar multiplication:* For $\alpha, \beta \in \mathbb{C}$ and $f \in \mathcal{C}[a, b]$,

$$((\alpha\beta)f)(x) = (\alpha\beta)f(x) = \alpha(\beta f(x)) = (\alpha(\beta f))(x).$$

- *Identity scalar:* For $f \in \mathcal{C}[a, b]$,

$$(1 \cdot f)(x) = 1 \cdot f(x) = f(x),$$

hence $1 \cdot f = f$.

- *Distributivity over function addition:* For $\alpha \in \mathbb{C}$ and $f, g \in \mathcal{C}[a, b]$,

$$(\alpha(f + g))(x) = \alpha(f(x) + g(x)) = \alpha f(x) + \alpha g(x) = (\alpha f + \alpha g)(x).$$

- *Distributivity over scalar addition:* For $\alpha, \beta \in \mathbb{C}$ and $f \in \mathcal{C}[a, b]$,

$$((\alpha + \beta)f)(x) = (\alpha + \beta)f(x) = \alpha f(x) + \beta f(x) = (\alpha f + \beta f)(x).$$

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All vector space axioms over $\mathbb{C}$ are therefore satisfied, so $\mathcal{C}[a, b]$ is a complex vector space. We first check that $\langle \cdot, \cdot \rangle$ satisfies the axioms of an inner product on a complex vector space.

(1) Conjugate symmetry. For any $u, v \in \mathcal{C}[a, b]$ we have

$$\overline{\langle v, u \rangle} = \overline{\int_a^b \rho(t) v(t) \overline{u(t)} dt} = \int_a^b \rho(t) \overline{v(t)} u(t) dt = \int_a^b \rho(t) u(t) \overline{v(t)} dt = \langle u, v \rangle.$$

(2) Linearity in the first slot. Let $u_1, u_2, v \in \mathcal{C}[a, b]$ and $\alpha, \beta \in \mathbb{C}$. Using linearity of the (Lebesgue or Riemann) integral, we obtain

$$\begin{aligned} \langle \alpha u_1 + \beta u_2, v \rangle &= \int_a^b \rho(t) (\alpha u_1(t) + \beta u_2(t)) \overline{v(t)} dt \\ &= \alpha \int_a^b \rho(t) u_1(t) \overline{v(t)} dt + \beta \int_a^b \rho(t) u_2(t) \overline{v(t)} dt \\ &= \alpha \langle u_1, v \rangle + \beta \langle u_2, v \rangle. \end{aligned}$$

(3) Conjugate linearity in the second slot. This follows from (1) and (2). Indeed, for $u, v_1, v_2 \in \mathcal{C}[a, b]$ and $\alpha, \beta \in \mathbb{C}$,

$$\begin{aligned} \langle u, \alpha v_1 + \beta v_2 \rangle &= \overline{\langle \alpha v_1 + \beta v_2, u \rangle} \\ &= \overline{\alpha \langle v_1, u \rangle + \beta \langle v_2, u \rangle} \\ &= \overline{\alpha} \overline{\langle v_1, u \rangle} + \overline{\beta} \overline{\langle v_2, u \rangle} \\ &= \overline{\alpha} \langle u, v_1 \rangle + \overline{\beta} \langle u, v_2 \rangle. \end{aligned}$$

(4) Positive definiteness. For any $u \in \mathcal{C}[a, b]$,

$$\langle u, u \rangle = \int_a^b \rho(t) |u(t)|^2 dt.$$

Since $\rho(t) > 0$ and $|u(t)|^2 \geq 0$ for all t , the integrand is non-negative, hence $\langle u, u \rangle \geq 0$.

If $\langle u, u \rangle = 0$, then $\int_a^b \rho(t) |u(t)|^2 dt = 0$. Because the integrand is continuous and non-negative and $\rho(t) > 0$, this can happen only if $\rho(t) |u(t)|^2 = 0$ for all $t \in [a, b]$. Thus $|u(t)|^2 = 0$ for all t , so $u(t) = 0$ for all $t \in [a, b]$, that is, $u \equiv 0$. Conversely, if $u \equiv 0$, then clearly $\langle u, u \rangle = 0$. Hence $\langle u, u \rangle = 0$ if and only if $u = 0$.

Combining (1)–(4), we see that $\langle \cdot, \cdot \rangle$ is indeed an inner product on $\mathcal{C}[a, b]$.

Now define

$$\|u\|_2 := \sqrt{\langle u, u \rangle} = \left(\int_a^b \rho(t) |u(t)|^2 dt \right)^{1/2}.$$

On any inner-product space the function $\|u\| := \sqrt{\langle u, u \rangle}$ satisfies the three norm axioms: $\|u\| \geq 0$ and $\|u\| = 0$ iff $u = 0$, $\|\alpha u\| = |\alpha| \|u\|$, and the triangle inequality $\|u + v\| \leq \|u\| + \|v\|$, the last one following from the Cauchy–Schwarz inequality. Therefore $\|\cdot\|_2$ is a norm on $\mathcal{C}[a, b]$.

Hence $\mathcal{C}[a, b]$ equipped with $\langle \cdot, \cdot \rangle$ is an inner-product space over $\mathbb{C}$, and the corresponding norm $\|\cdot\|_2$ makes $\mathcal{C}[a, b]$ into a normed vector space.

II. Chebyshev polynomials of the first kind

(a)

It suffices to show that for $m \neq n$ with $m, n \in \mathbb{N}$,

$$\int_{-1}^1 \frac{T_m(x) T_n(x)}{\sqrt{1-x^2}} dx = 0.$$

Using $T_k(x) = \cos(k \arccos x)$, we have

$$\int_{-1}^1 \frac{T_m(x) T_n(x)}{\sqrt{1-x^2}} dx = \int_{-1}^1 \frac{\cos(m \arccos x) \cos(n \arccos x)}{\sqrt{1-x^2}} dx.$$

Let $t = \arccos x$, so that $x = \cos t$ and $dx = -\sin t dt$. Since $\sqrt{1-x^2} = \sqrt{1-\cos^2 t} = \sin t$ for $t \in [0, \pi]$, we obtain

$$\int_{-1}^1 \frac{\cos(m \arccos x) \cos(n \arccos x)}{\sqrt{1-x^2}} dx = \int_{\pi}^0 \frac{\cos(mt) \cos(nt)}{\sin t} (-\sin t) dt = \int_0^{\pi} \cos(mt) \cos(nt) dt.$$

Now use the identity $\cos a \cos b = \frac{1}{2}(\cos(a+b) + \cos(a-b))$:

$$\int_0^\pi \cos(mt) \cos(nt) dt = \frac{1}{2} \int_0^\pi (\cos((m+n)t) + \cos((m-n)t)) dt.$$

If $m \neq n$, then $m+n \neq 0$ and $m-n \neq 0$, hence

$$\int_0^\pi \cos(kt) dt = \left[\frac{\sin(kt)}{k} \right]_0^\pi = 0 \quad (k \in \mathbb{Z} \setminus \{0\}),$$

so the integral is 0, as desired.

(b)

The first three Chebyshev polynomials of the first are

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_2(x) = 2x^2 - 1.$$

Using the result in (a)

$$\langle T_0, T_0 \rangle = \pi, \quad \langle T_n, T_n \rangle = \frac{\pi}{2} \quad (n \geq 1),$$

we get

$$\|T_0\| = \sqrt{\pi}, \quad \|T_1\| = \sqrt{\frac{\pi}{2}}, \quad \|T_2\| = \sqrt{\frac{\pi}{2}}.$$

Hence an orthonormal system is

$$\phi_0(x) = \frac{T_0(x)}{\|T_0\|} = \frac{1}{\sqrt{\pi}}, \quad \phi_1(x) = \frac{T_1(x)}{\|T_1\|} = \sqrt{\frac{2}{\pi}} x, \quad \phi_2(x) = \frac{T_2(x)}{\|T_2\|} = \sqrt{\frac{2}{\pi}} (2x^2 - 1).$$

Then $\langle \phi_i, \phi_j \rangle = \delta_{ij}$ for $i, j \in \{0, 1, 2\}$ Then is an orthonormal system.

III. Least-squares approximation of $y(x) = \sqrt{1-x^2}$ on $[-1, 1]$

We use the inner product from Theorem 5.7 with

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)\rho(x)dx, \quad \rho(x) = \frac{1}{\sqrt{1-x^2}}.$$

Let $f(x) = \sqrt{1-x^2}$. So $\langle f, g \rangle = \int_{-1}^1 g(x)dx$

(a) Fourier expansion

By result in II(b) $T_n(\cos t) = \cos(nt)$.

$$\phi_0(x) = \frac{T_0(x)}{\|T_0\|} = \frac{1}{\sqrt{\pi}}, \quad \phi_1(x) = \frac{T_1(x)}{\|T_1\|} = \sqrt{\frac{2}{\pi}} x, \quad \phi_2(x) = \frac{T_2(x)}{\|T_2\|} = \sqrt{\frac{2}{\pi}} (2x^2 - 1).$$

Compute the coefficients:

$$a_0 = \langle f, \phi_0 \rangle = \int_{-1}^1 \phi_0(x) dx = \int_{-1}^1 \frac{1}{\sqrt{\pi}} dx = \frac{2}{\sqrt{\pi}},$$

$$a_1 = \langle f, \phi_1 \rangle = \int_{-1}^1 \phi_1(x) dx = \int_{-1}^1 \sqrt{\frac{2}{\pi}} x dx = 0,$$

$$a_2 = \langle f, \phi_2 \rangle = \int_{-1}^1 \phi_2(x) dx = \int_{-1}^1 \sqrt{\frac{2}{\pi}} (2x^2 - 1) dx = \sqrt{\frac{2}{\pi}} \left(2 \cdot \frac{2}{3} - 2 \right) = -\frac{2}{3} \sqrt{\frac{2}{\pi}}.$$

Thus

$$p_2(x) = \frac{2}{\sqrt{\pi}} \cdot \frac{1}{\sqrt{\pi}} + \left(-\frac{2}{3} \sqrt{\frac{2}{\pi}} \right) \cdot \sqrt{\frac{2}{\pi}} (2x^2 - 1) = \frac{2}{\pi} - \frac{4}{3\pi} (2x^2 - 1) = \frac{10}{3\pi} - \frac{8}{3\pi} x^2.$$

(b) normal equations

Let $p_2(x) = c_0 + c_1x + c_2x^2$. Then we know

$$\langle f - p_2, 1 \rangle = 0, \quad \langle f - p_2, x \rangle = 0, \quad \langle f - p_2, x^2 \rangle = 0,$$

Since $f(x)$ and $\rho(x) = \frac{1}{\sqrt{1-x^2}}$ are even, while x is odd, we have $\langle f, x \rangle = 0$, $\langle 1, x \rangle = 0$, and $\langle x^2, x \rangle = 0$. Thus $\langle f - p_2, x \rangle = 0$ forces $c_1 = 0$.

Now compute the needed inner products directly:

$$\langle f, 1 \rangle = \int_{-1}^1 1 dx = 2,$$

$$\langle 1, 1 \rangle = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx = \pi, \quad \langle x^2, 1 \rangle = \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{2}.$$

Hence $\langle f - p_2, 1 \rangle = 0$ becomes

$$2 - c_0\pi - c_2\frac{\pi}{2} = 0.$$

Then

$$\langle f, x^2 \rangle = \int_{-1}^1 x^2 dx = \frac{2}{3},$$

$$\langle 1, x^2 \rangle = \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{2}, \quad \langle x^2, x^2 \rangle = \int_{-1}^1 \frac{x^4}{\sqrt{1-x^2}} dx = \frac{3\pi}{8}.$$

Therefore $\langle f - p_2, x^2 \rangle = 0$ becomes

$$\frac{2}{3} - c_0\frac{\pi}{2} - c_2\frac{3\pi}{8} = 0.$$

Solving the equation

$$c_2 = -\frac{8}{3\pi}, \quad c_0 = \frac{10}{3\pi}, \quad c_1 = 0,$$

hence

$$p_2(x) = \frac{10}{3\pi} - \frac{8}{3\pi}x^2,$$

IV. DLS

data. We have $N = 12$, $\rho \equiv 1$, and the sales record (Example 5.55) is

$$(x_i, y_i) = (1, 256), (2, 201), (3, 159), (4, 61), (5, 77), (6, 40), (7, 17), (8, 25), (9, 103), (10, 156), (11, 222), (12, 345).$$

Define the discrete inner product

$$\langle u, v \rangle = \sum_{i=1}^{12} u(x_i)v(x_i).$$

IV(a) Orthonormal polynomials via Gram–Schmidt

Start from the independent list $(1, x, x^2)$.

Step 1. Let $p_0(x) = 1$. Then

$$\langle p_0, p_0 \rangle = \sum_{i=1}^{12} 1 = 12, \quad \phi_0(x) = \frac{p_0(x)}{\|p_0\|} = \frac{1}{\sqrt{12}}.$$

Step 2. Let $v_1(x) = x$. Remove its projection onto ϕ_0 :

$$\langle v_1, \phi_0 \rangle = \sum_{i=1}^{12} x_i \cdot \frac{1}{\sqrt{12}} = \frac{1}{\sqrt{12}} \sum_{i=1}^{12} i = \frac{78}{\sqrt{12}}.$$

Hence

$$p_1(x) = v_1(x) - \langle v_1, \phi_0 \rangle \phi_0(x) = x - \frac{78}{12} = x - \frac{13}{2}.$$

Moreover,

$$\langle p_1, p_1 \rangle = \sum_{i=1}^{12} \left(i - \frac{13}{2} \right)^2 = 143, \quad \phi_1(x) = \frac{p_1(x)}{\|p_1\|} = \frac{x - \frac{13}{2}}{\sqrt{143}}.$$

Step 3. Let $v_2(x) = x^2$. Remove its projections onto ϕ_0, ϕ_1 and set

$$p_2(x) = v_2(x) - \langle v_2, \phi_0 \rangle \phi_0(x) - \langle v_2, \phi_1 \rangle \phi_1(x).$$

A direct computation gives

$$p_2(x) = x^2 - 13x + \frac{91}{3}.$$

Furthermore,

$$\langle p_2, p_2 \rangle = \sum_{i=1}^{12} \left(i^2 - 13i + \frac{91}{3} \right)^2 = \frac{4004}{3}.$$

Equivalently, multiplying by 3 to avoid fractions,

$$\phi_2(x) = \frac{3p_2(x)}{\|3p_2\|} = \frac{3x^2 - 39x + 91}{\sqrt{12012}},$$

since $\langle 3p_2, 3p_2 \rangle = 9 \cdot \frac{4004}{3} = 12012$.

Therefore an orthonormal system is

$$\phi_0(x) = \frac{1}{\sqrt{12}}, \quad \phi_1(x) = \frac{x - \frac{13}{2}}{\sqrt{143}}, \quad \phi_2(x) = \frac{3x^2 - 39x + 91}{\sqrt{12012}}.$$

IV(b) Best quadratic least-squares approximation

We seek $\hat{\varphi}(x) = a_0 + a_1x + a_2x^2$ minimizing

$$\sum_{i=1}^{12} (y_i - \hat{\varphi}(x_i))^2 = \|y - \hat{\varphi}\|^2$$

with respect to the discrete inner product.

Using the orthonormal basis. Write

$$\hat{\varphi}(x) = c_0\phi_0(x) + c_1\phi_1(x) + c_2\phi_2(x), \quad c_k = \langle y, \phi_k \rangle.$$

Compute

$$\begin{aligned} c_0 &= \sum_{i=1}^{12} y_i \cdot \frac{1}{\sqrt{12}} = \frac{\sum_{i=1}^{12} y_i}{\sqrt{12}} = \frac{1662}{\sqrt{12}}, \\ c_1 &= \sum_{i=1}^{12} y_i \cdot \frac{x_i - \frac{13}{2}}{\sqrt{143}} = \frac{\sum_{i=1}^{12} y_i (x_i - \frac{13}{2})}{\sqrt{143}} = \frac{589}{\sqrt{143}}, \\ c_2 &= \sum_{i=1}^{12} y_i \cdot \frac{3x_i^2 - 39x_i + 91}{\sqrt{12012}} = \frac{\sum_{i=1}^{12} y_i (3x_i^2 - 39x_i + 91)}{\sqrt{12012}} = \frac{36204}{\sqrt{12012}}. \end{aligned}$$

Converting back to the monomial basis yields

$$\hat{\varphi}(x) = 386 - \frac{16220}{143}x + \frac{1293}{143}x^2$$

(i.e. $\hat{\varphi}(x) \approx 386 - 113.4265734x + 9.041958042x^2$).

(Verification via normal equations.) Let $A = [1, x, x^2] \in \mathbb{R}^{12 \times 3}$ with rows $(1, x_i, x_i^2)$. Then the normal equations are

$$(A^\top A)a = A^\top y, \quad a = (a_0, a_1, a_2)^\top.$$

Here

$$A^\top A = \begin{pmatrix} 12 & 78 & 650 \\ 78 & 650 & 6084 \\ 650 & 6084 & 60710 \end{pmatrix}, \quad A^\top y = \begin{pmatrix} 1662 \\ 11392 \\ 109750 \end{pmatrix},$$

whose solution is exactly

$$a_0 = 386, \quad a_1 = -\frac{16220}{143}, \quad a_2 = \frac{1293}{143},$$

agreeing with the result above.

IV(c) Reuse when y_i changes but x_i and N stay the same

If another sales record has the same nodes x_i and N (and the same weight $\rho \equiv 1$) but different values y_i , then:

- The orthonormal polynomials ϕ_0, ϕ_1, ϕ_2 from part (a) can be reused, since they depend only on $\{x_i\}$ and ρ .
- Only the coefficients $c_k = \langle y, \phi_k \rangle = \sum_{i=1}^{12} y_i \phi_k(x_i)$ need to be recomputed.

This reuse illustrates an advantage of orthonormal polynomials over normal equations: the (expensive) orthogonalization can be done once, and subsequent fits for new y reduce to computing a few inner products, with better numerical stability than solving normal equations repeatedly.

V. New method of rounding off number

By the definition in the course, the unit roundoff represents the maximum round off error around 1, where the floating number is the densest. By the new method of rounding off number : the badest case is the number around 1 is $1 + 2^{-23}$ in IEEE-754. So

$$\epsilon_u^{new} = 2^{-23} = 2\epsilon_u$$

VI. Estimate the precision lost in the subtraction

We know:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + O(x^6)$$

$\Rightarrow$

$$1 - \cos \frac{1}{4} = 2^{-5} + \epsilon \quad (\epsilon \ll 2^{-5})$$

So the subtraction will lose 5 bits of precision.

VII. Symmetric Polynomials

Method 1:

$$1 - \cos x = 2 \sin^2\left(\frac{x}{2}\right)$$

Then we know near 0 $2 \sin^2(\frac{x}{2})$ is small. So without the subtraction of two closed number. It will NOT cause a catastrophic cancellation.

Method 2:

$$1 - \cos x = \frac{1 - \cos^2 x}{1 + \cos x} = \frac{\sin^2 x}{1 + \cos x}$$

For the denominator, both of them are the number closed to 1, which will NOT cause a catastrophic cancellation. Meanwhile for the numerator, it's just a square of small number, which also avoid a subtraction of two closed number.

VIII. Condition numbers of functions

We know $\text{cond}_f(x) = \left| \frac{f'(x)x}{f(x)} \right|$

VIII.a

When $\alpha = 0$ the function is trivial. We only consider the case $\alpha \neq 0$

$$\text{cond}_f(x) = \left| \frac{\alpha(x-1)^{\alpha-1}x}{(x-1)^\alpha} \right| = \left| \frac{\alpha x}{x-1} \right|$$

So the condition number is large around $x = 1$.

VIII.b

$$\text{cond}_f(x) = \left| \frac{\frac{1}{x}x}{\ln x} \right| = \left| \frac{1}{\ln x} \right|$$

So the condition number is large around $x = 1$.

VIII.c

$$\text{cond}_f(x) = \left| \frac{e^x x}{e^x} \right| = |x|$$

So the condition number is large when $x \rightarrow \pm\infty$.

VIII.d

$$\text{cond}_f(x) = \left| \frac{x \frac{-1}{\sqrt{1-x^2}}}{\arccos x} \right| = \left| \frac{x}{\arccos x \sqrt{1-x^2}} \right|$$

So the condition number is large around $x = \pm 1$.

IX. Function $f(x) = 1 - e^{-x}$

IX.a

$$\text{cond}_f(x) = \left| \frac{f'(x)x}{f(x)} \right| = \left| \frac{x e^{-x}}{1 - e^{-x}} \right| = \left| \frac{x}{e^x - 1} \right|$$

When $x \in (0, 1]$, $\text{cond}_f(x) = \frac{x}{e^x - 1}$, besides we know $e^x \geq 1 + x$. Meanwhile, $\lim_{x \rightarrow 0} \frac{x}{e^x - 1} = 1$. So $\text{cond}_f(x) \leq 1$, when $x \in [0, 1]$.

IX.b

We know $f_A(x) = (1 - (1 + \delta_1)e^{-x})(1 + \delta_2)$, and $\delta_1, \delta_2 \leq \epsilon_u$, so

$$|\delta(x)| = \left| \frac{f_A(x) - f(x)}{f(x)} \right| = \left| \frac{\delta_2 - (\delta_1 + \delta_2 + \delta_1 \delta_2)e^{-x}}{1 - e^{-x}} \right| \leq \left| \frac{\delta_2 + (\delta_1 + \delta_2 + \delta_1 \delta_2)e^{-x}}{1 - e^{-x}} \right| \approx \left| \frac{\epsilon_u e^x + 2\epsilon_u}{e^x - 1} \right| = \epsilon_u \left| \frac{e^x + 2}{e^x - 1} \right|$$

Therefore:

$$\text{cond}_A(x) \leq \frac{|\delta(x)|}{\epsilon_u \text{cond}_f(x)} = \frac{|e^x + 2|}{|x|}$$

when $x \rightarrow 0$, the condition number $\text{cond}_A(x) \rightarrow \infty$, it's ill. In the other points, the algorithm works.

X. Prove a lemma

We know

$$\|A\|_2 = \max_{\|x\|=1} \|Ax\|_2 = \max_{\|x\|=1} \sqrt{x^* A^* A x}$$

Attention $A^* A$ is symmetric matrix, which can be diagonalized. Let $\{v_i\}_{1 \leq i \leq n}$ be the eigen vectors of $\{\mu_i\}_{1 \leq i \leq n}$ of $A^* A$ and o.n.b by the way (eigen vectors in different eigen vector space are orthogonal). Then every x can be written

as $\sum_{i=1}^n \alpha_i v_i$, so

$$\|A\|_2 = \max_{\|x\|=1} \sqrt{x^* A^* A x} = \max_{\sum_{i=1}^n \alpha_i^2 = 1} \sqrt{\sum_{i=1}^n \mu_i^2 \alpha_i^2} < v_i, v_i > \leq \sigma_{\max}$$

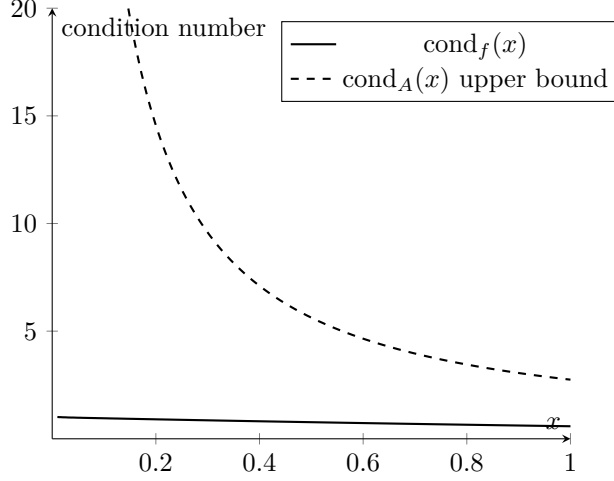


Figure 1: Condition numbers on $[0, 1]$

Meanwhile

$$\|A\|_2 \geq \max_{1 \leq i \leq n} \sqrt{v_i^* A^* A v_i} = \sigma_{max}$$

Then $\|A\|_2 = \sigma_{max}$. And notice that $(A^{-1})^* A^{-1} = (AA^*)^{-1}$. Besides by linear algebra $|\lambda E - AB| = |\lambda E - BA|$. So we know AA^* has the same eigen value as A^*A . Then $\|A^{-1}\|_2 = \frac{1}{\sigma_{min}}$, since the eigen value of $(AA^*)^{-1}$ is $\{\frac{1}{\mu_i}\}_{1 \leq i \leq n}$. So $\text{cond}_2 A = \frac{\sigma_{max}}{\sigma_{min}}$.

If $A = A^*$, then $\exists$ unitary matrix U , s.t.

$$UAU^* = \text{diag}(\lambda_1, \dots, \lambda_n) \Rightarrow UA^*AU^* = UAU^*UAU^* = \text{diag}(\lambda_1^2, \dots, \lambda_n^2)$$

Then $\text{cond}_2 A = \frac{|\lambda_{max}|}{|\lambda_{min}|}$.

If $A^*A = E \Rightarrow \mu_i = 1, \forall i$, so $\sigma_{max} = \sigma_{min} = 1$. Then $\text{cond}_2 A = 1$.

XI. Find root of a polynomial

We know

$$q(r) = 0 \Rightarrow r^n + \sum_{i=0}^{n-1} a_i r^i = 0 \Rightarrow \frac{\partial r}{\partial a_i} q'(r) + r^i = 0$$

$$\text{cond}_f(\mathbf{x}) = \|A(\mathbf{x})\|_1 = \|(a_{ij}(\mathbf{x}))\|_1 = \max_{0 \leq i \leq n-1} \left| \frac{a_i \frac{\partial r}{\partial a_i}}{r} \right| = \max_{0 \leq i \leq n-1} \left| \frac{a_i r^{i-1}}{q'(r)} \right|$$

Besides for root r

$$W(x) := \sum_{i=1}^m (x - i) \Rightarrow \text{cond}_f(\mathbf{x}) = \max_{0 \leq i \leq m-1} \left| \frac{r^{i-1} a_i}{W'(r)} \right| = \max_{0 \leq i \leq m-1} \left| \frac{r^{i-1} a_i}{\prod_{j=1, j \neq r}^m (r - j)} \right|$$

Specially $r = m = 20$, $\text{cond}_f(\mathbf{x}) \approx 2.0 \times 10^{10}$.

$r = m = 30$, $\text{cond}_f(\mathbf{x}) \approx 2.0 \times 10^{16}$.

$r = m = 40$, $\text{cond}_f(\mathbf{x}) \approx 1.5 \times 10^{22}$.

Then by comparison, i see that if a question's condition is large enough, it's almost impossible to find a numerical algorithm that make the condition small.

XII. Given a counter example

Choose $\beta = 10, p = 2$. Then given $1/(2/7)$.

$$fl_p(1/fl_p(2/7)) = fl_p(1/0.29) = 3.4$$

$$fl_{2p}(1/fl_{2p}(2/7)) = fl_{2p}(1/0.2857) = 3.500 \rightarrow 3.5$$

Then get a contradiction.

XIII. Accuracy of computing root in IEEE 754

The answer is NO.

In IEEE 754 $\beta = 2, p = 24, L = -126, U = 127$. We know at least after $\lceil \frac{6}{\lg 2} \rceil + 1 = 20$ steps, we acquire the 10^{-6} accuracy. But the number $\in [128, 129]$, expressed as $(1.a_1 \dots a_{23})_2 \times 2^7$. The accuracy of the number is at most $2^{7-23-1} = 2^{-17}$, which at most ensure the accuracy of 17 steps. So we can't compute the root with absolute accuracy $< 10^{-6}$.

XIV. Explain a phenomenon

We know, for a pair every close adjacent points, may as well $x_1 = x_0 + h$, in this interval:

$$s(x) = a(x - x_0)^3 + b(x - x_0)^2 + c(x - x_0) + d$$

So we have

$$S[a, b, c, d]^T = M$$

For the curve is continuous, we know there are two line elements of S are

$$(0, 0, 0, 1) \quad (h^3, h^2, h, 1)$$

which is almost linear dependent when $h \rightarrow 0$, so the $\sigma_{\min} \rightarrow \infty$. Then the condition of S , $\text{cond}_2(S) \rightarrow \infty$, which will make the problem ill. Therefore will get an inaccurate result.